

• DATA 202 • FINAL PRESENTATION

Statistical Analysis of Customer Spending at Starbucks.

*"Every great solution starts
with a good cup of coffee."*

PRESENTED BY

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A behavioral map of 100,000 Starbucks orders.

Why customer spend matters — and what we can ask of this data.

100,000

orders

Order-level transactions, no missing values

14,988

unique customers

Repeat-purchase history captured

500

stores

Urban, suburban & rural locations

20

variables

Channel, cart, customizations, satisfaction...

WHY THIS MATTERS

Spend per order is the single revenue lever that ties together channel strategy, loyalty design, and the in-store experience.

With zero missing values and twenty rich behavioral variables, this dataset lets us go beyond averages to ask what actually *drives* a higher ticket.

What factors drive customer spending at Starbucks?

A central question, broken into five DATA 202 lenses.

01 Which segments spend more? Channel, rewards, age, store type.

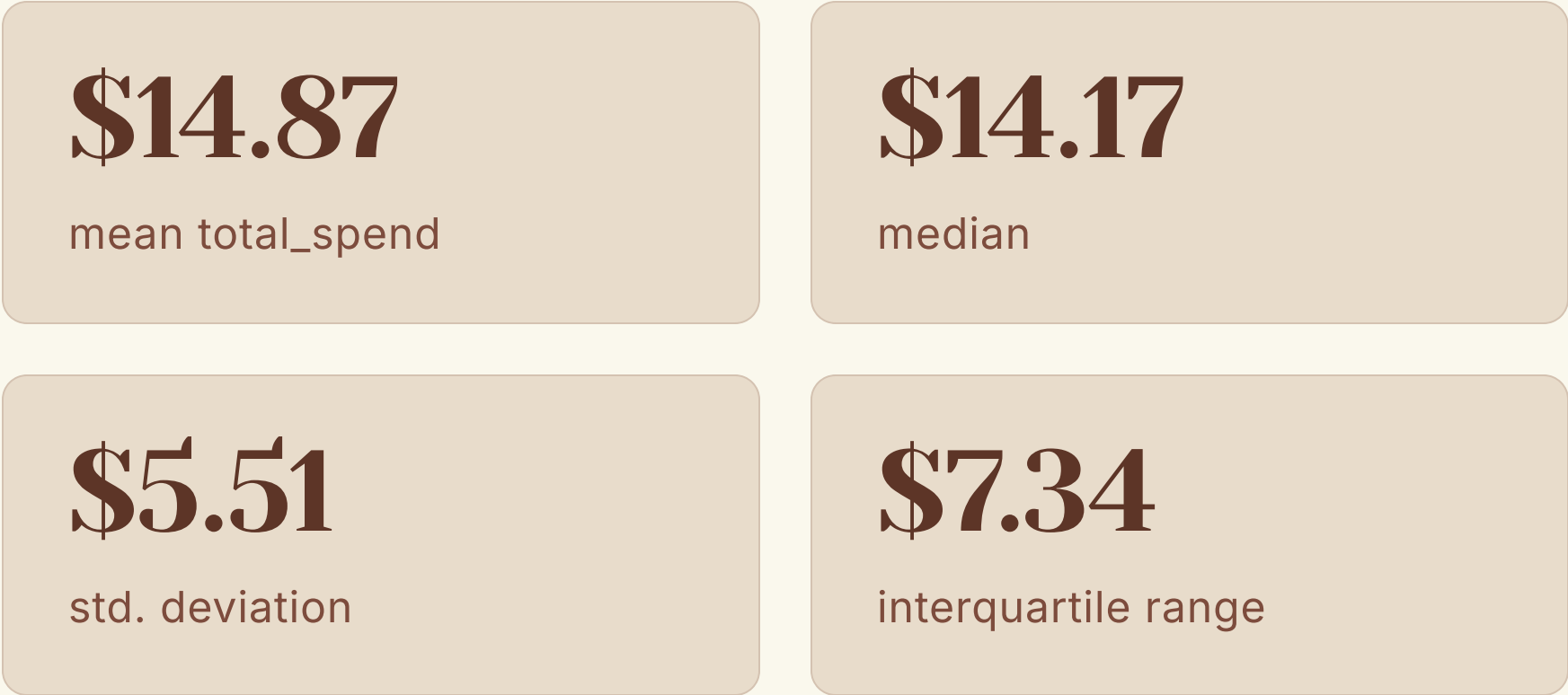
02 How do cart size and customizations relate to spend? Correlation & covariance.

03 Are Mobile and Rewards differences statistically real? Two-sample & paired tests.

04 How likely are high-spender orders — and who places them? Probability & Bayes.

05 Can linear algebra and optimization model spend? Normal equations & gradient descent.

The shape of an average order.



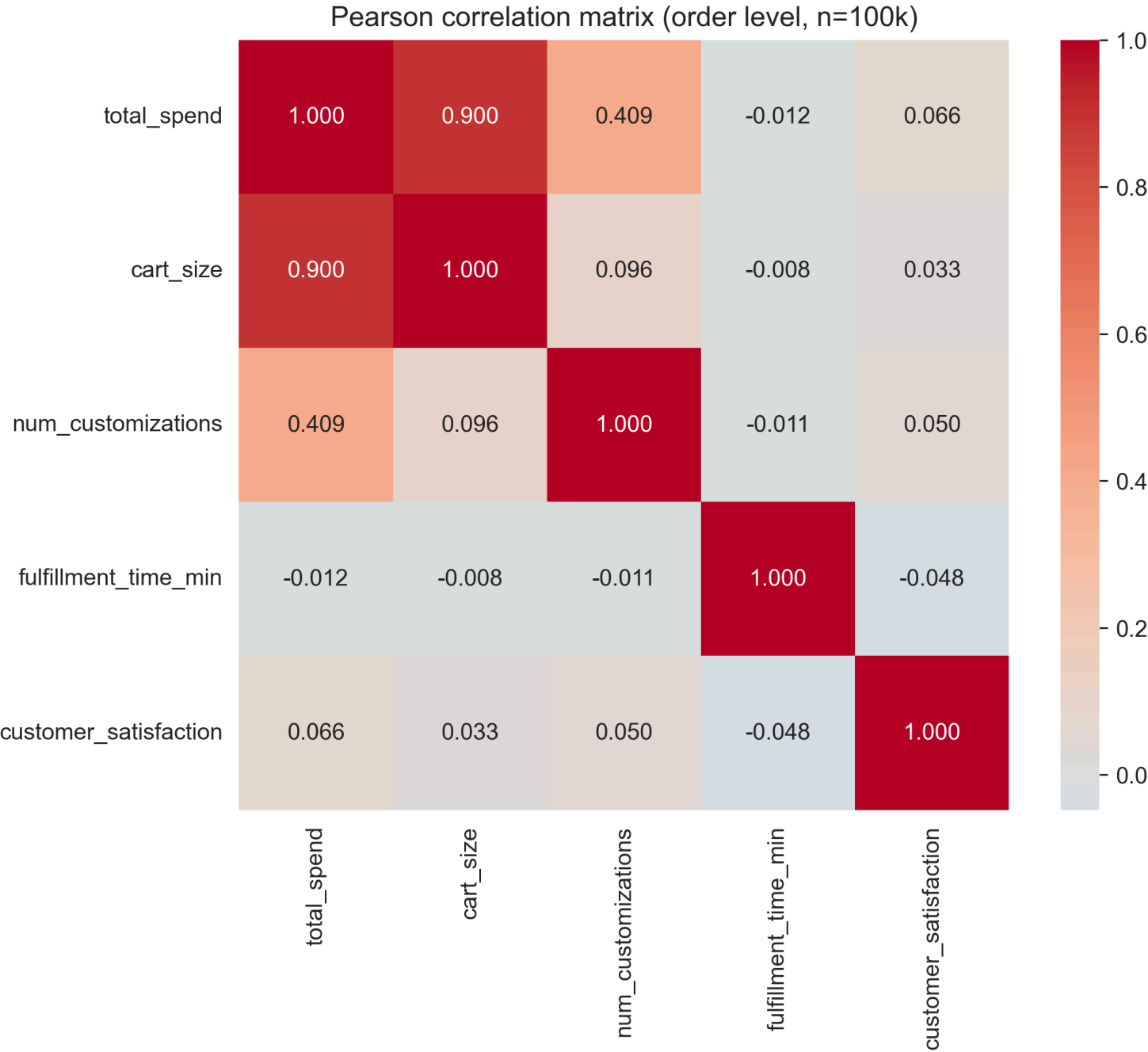
Outliers above \$29.19 · just 1.42% of orders.

MEAN SPEND BY SEGMENT

Segment	Mean spend
Mobile App	\$18.08
Drive-Thru	\$12.48
In-Store Cashier	\$12.51
Kiosk	\$12.49
Rewards member	\$15.72
Non-rewards	\$14.09

Mobile App leads every other channel by ~\$5.58.

Cart size, not customizations, is the dominant driver.



cart_size  **r = 0.900**

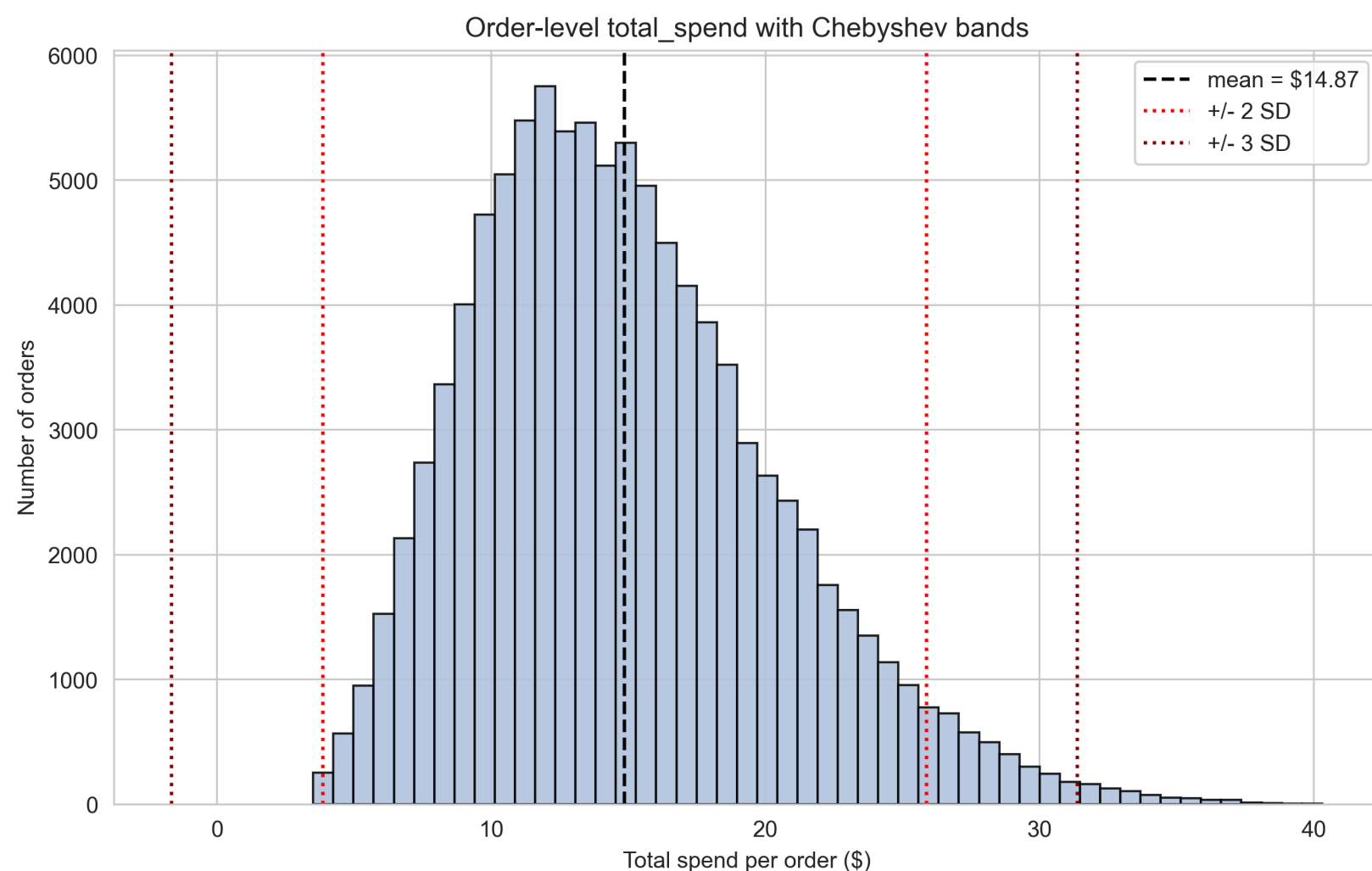
num_customizations  **r = 0.409**

customer_satisfaction  **r = 0.066**

fulfillment_time_min  **r = -0.012**

Cart size is near-deterministic for spend. Customizations matter, but only secondarily. Fulfillment time and satisfaction are essentially uncorrelated with spend.

Tighter than Chebyshev, but not quite normal.



95.83%

of orders within ± 2 SD

Empirical

99.34%

within ± 3 SD

Empirical

$\geq 75\%$

Chebyshev floor at 2 SD

Distribution-free

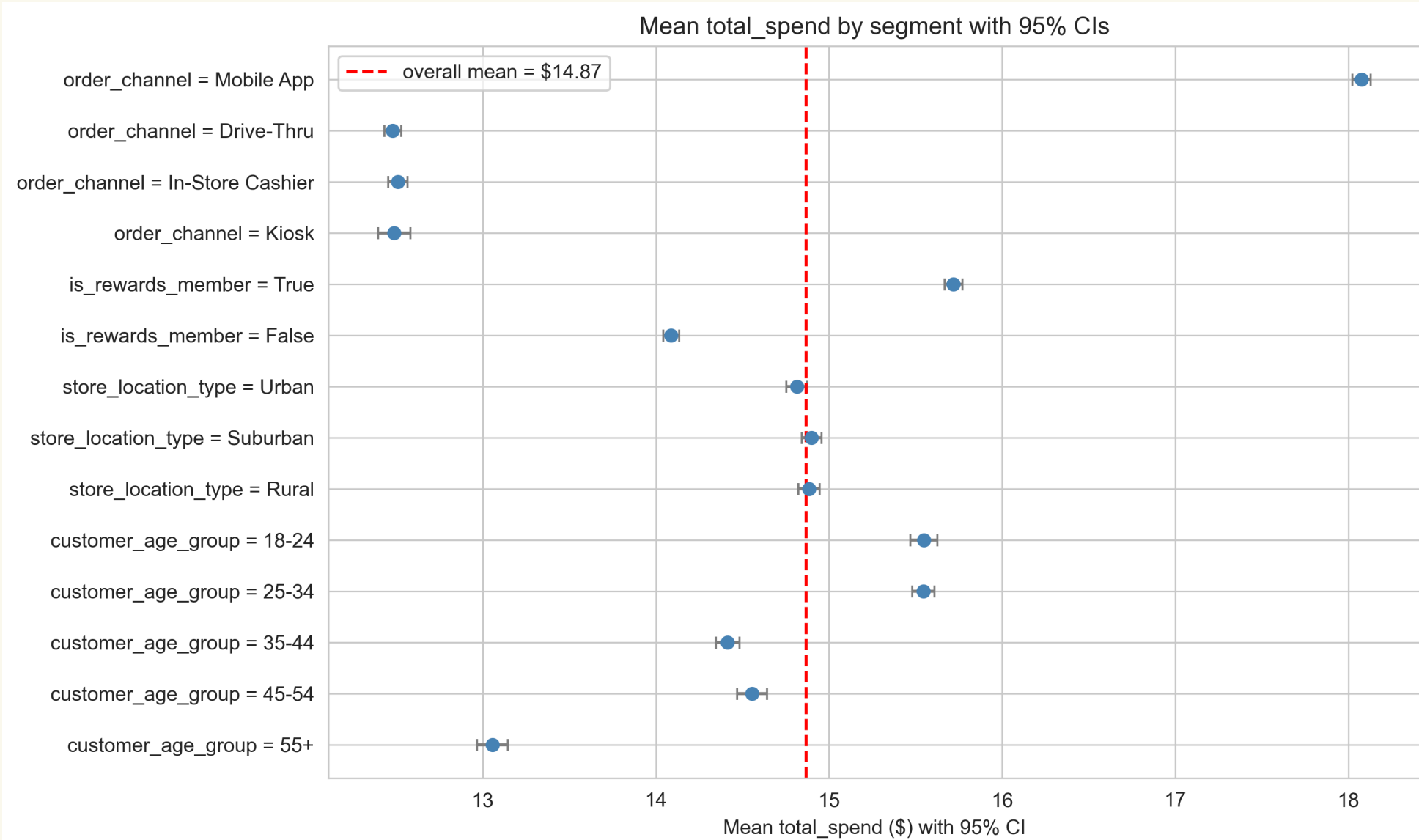
right-skew

moderate

Normal underestimates the
right tail

Because normality is imperfect, we use **empirical probabilities** — not the normal approximation — for tail events.

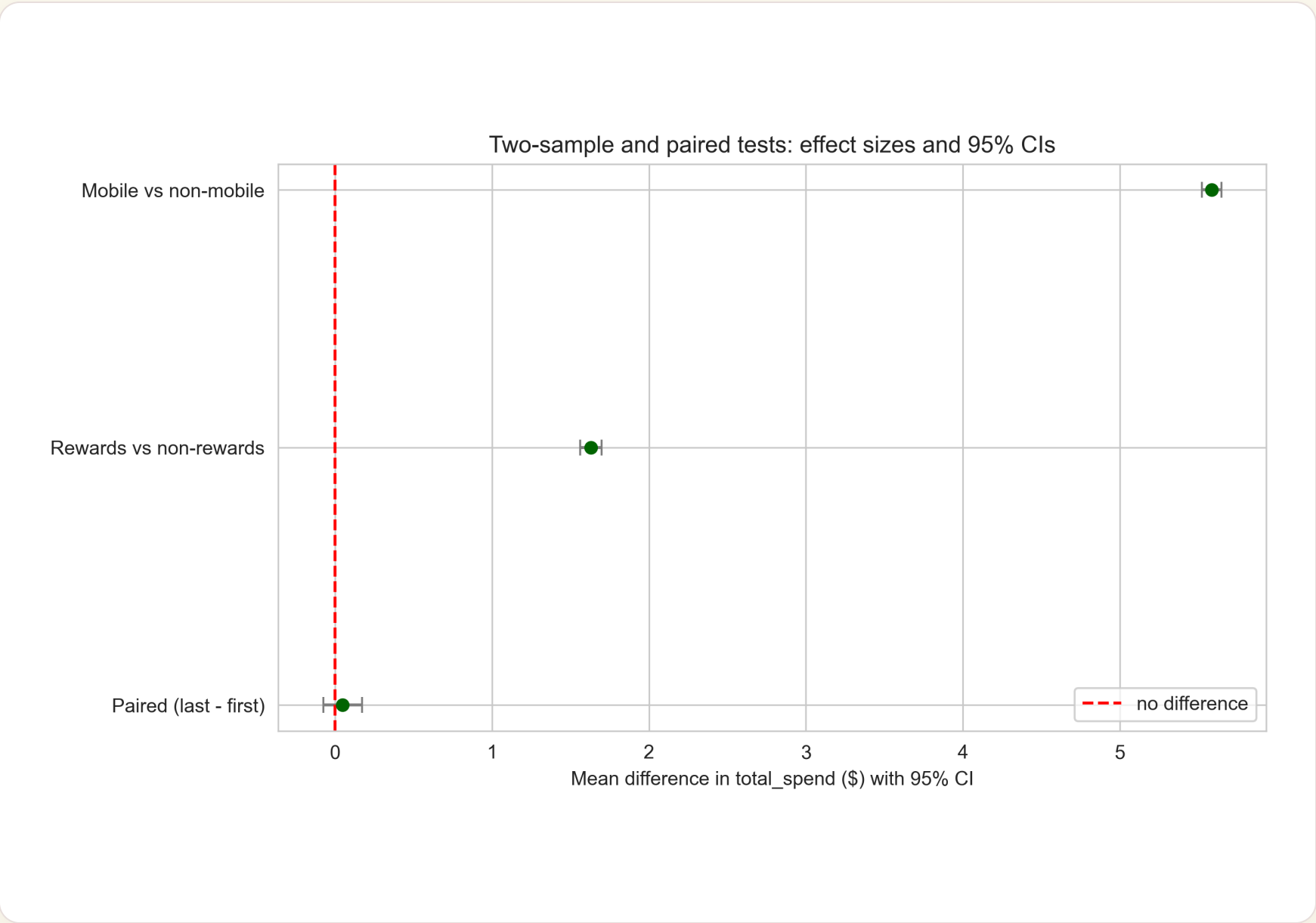
Mobile and Rewards bands don't overlap with the rest.



Segment	95% CI
Overall mean	[14.83, 14.90]
Mobile App	[18.02, 18.13]
Rewards	[15.67, 15.77]
Non-Rewards	[14.04, 14.13]

With $n \approx 100,000$, intervals are narrow. The Mobile band sits well above every other channel — previewing the hypothesis tests.

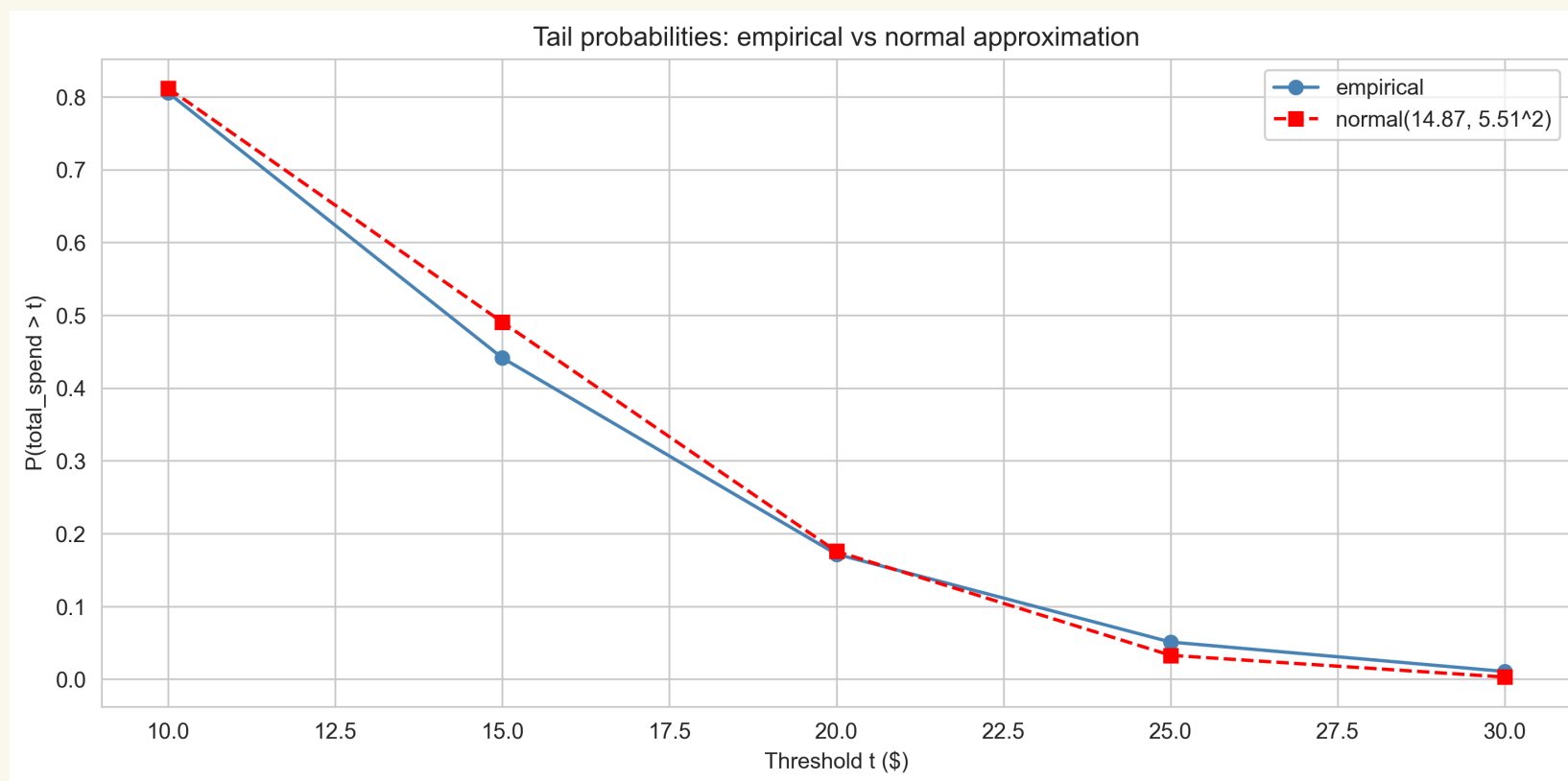
Four tests, four verdicts.



Test	Diff	Stat	p	Decision
Mobile vs Non-Mobile	\$5.58	t=175.91	≈ 0	REJECT H ₀
Rewards vs Non-Rewards	\$1.63	t=47.08	≈ 0	REJECT H ₀
First vs Latest (paired)	~ 0	—	0.44	FAIL TO REJECT
>\$20 high-spender Rewards vs Non	+8.6pp	z = 35.92	≈ 0	REJECT H ₀

Mobile and Rewards differences are **statistically significant**. Repeat-customer spend is **stable over time**.

High spenders are almost always Mobile.



5.11%

$P(\text{spend} > \$25)$ *empirical*

3.29%

Normal approximation
(underestimates tail)

11.27%

$P(\text{spend} > \$25 \mid \text{Mobile})$

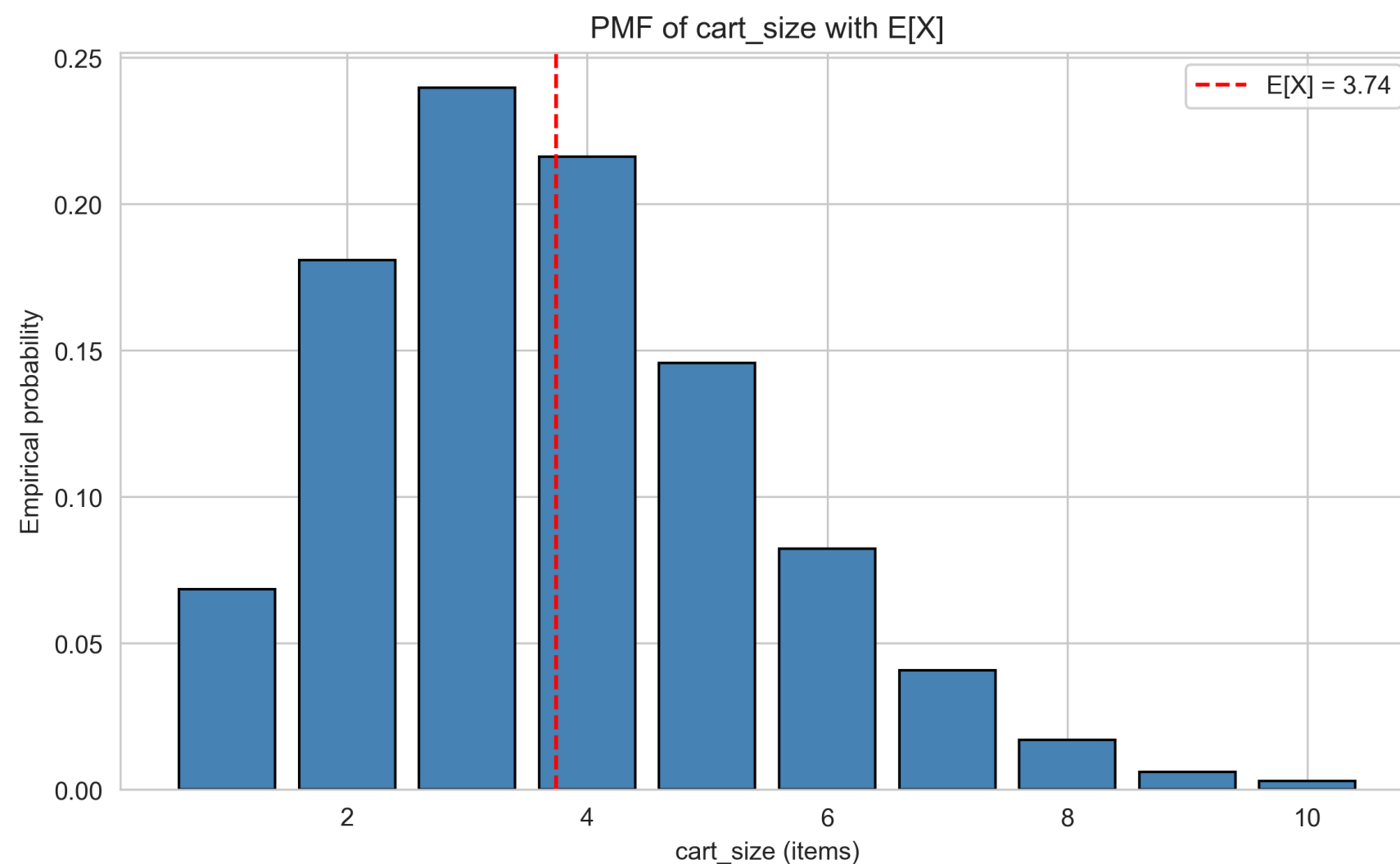
93.85%

$P(\text{Mobile} \mid \text{spend} > \$25)$
Bayes

And $P(\text{Rewards} \mid \text{spend} > \$25) = \mathbf{62.74\%}$ — high spenders are disproportionately Mobile, Rewards customers.

Cart size, formally.

A discrete RV over $\{1, 2, \dots, 10\}$ — the unit of work behind every dollar of spend.



3.7415

$E[\text{cart_size}]$ · expected items per order

2.8826

$\text{Var}(\text{cart_size})$

Mode at 3 items, right-skewed tail to 10. Treating cart size as an RV ties the descriptive picture directly to the linear model on the next slide.

A 3-feature model that explains 95% of spend.

$$\hat{y} = \beta_0 + \beta_1 \cdot \text{cart_size} + \beta_2 \cdot \text{customizations} + \beta_3 \cdot \text{mobile_app}$$

intercept (β_0) + 2.16

cart_size + 2.69 / item

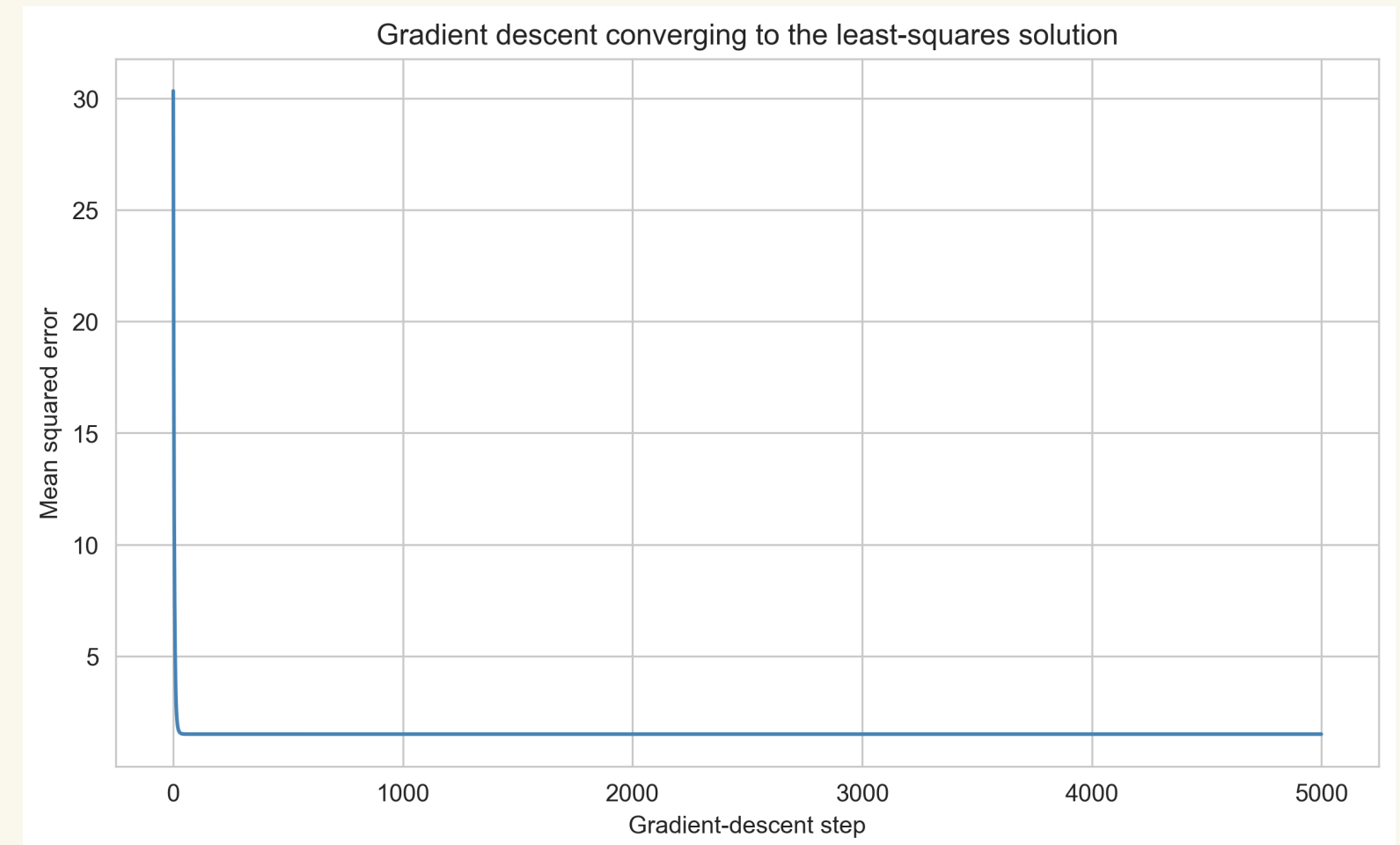
num_customizations + 0.92 / mod

mobile_app + 2.33

$R^2 = 0.9505$

95% of variance in total_spend explained

Solved two ways — closed-form via the normal equations, iteratively via gradient descent. Both converge to the same coefficients.



From coefficients to revenue levers.

Mobile App is the largest lever.

+\$5.58 vs every other channel. 93.85% of high-spender orders come through Mobile.

Cart size is the strongest predictor.

$r = 0.900$ with spend. Each extra item adds \$2.69 in the linear model.

Rewards is smaller but real.

+\$1.63 per order, statistically significant. Rewards members are 62.74% of high spenders.

High spenders are a Mobile phenomenon.

Channel mix — not just menu — is what unlocks above-\$25 orders.

From notebook to production.

01 Batch refresh

Recompute descriptive stats and refit the model on a regular cadence.

02 Online scoring

Predict expected spend at order time to drive personalized upsells.

03 Drift monitoring

Track Mobile share, Rewards effect, and high-spender probability over time.

04 Scale

Re-fit by store or region; same recipe, finer granularity.

The same three coefficients that explained 95% of variance in our notebook can power live recommendations in the Starbucks app.

Honest about what we can't claim.

- **Sample representativeness.** Kaggle data may not cover every real Starbucks store or geography.
- **Independence.** Repeat customers weaken strict independence assumptions in t-tests.
- **Normality testing.** Shapiro–Wilk uses a 5,000-row subsample because of its sample-size cap.
- **Tail estimation.** The Normal approximation underestimates the right tail of spend.
- **Paired test.** Compares first vs latest order, ignoring unequal time gaps between them.
- **Model simplicity.** Three predictors only — intentionally course-aligned, not state of the art.

Mobile App and cart size drive spend.

Four findings that hold across every method we ran.

- 01** **Mobile App and cart size** are the dominant drivers of spend per order.
- 02** **Rewards membership** is a smaller but statistically significant lift.
- 03** **Empirical tail probabilities** are more honest than the normal approximation for high spenders.
- 04** **Methods align with DATA 202** — descriptives, correlation, hypothesis tests, Bayes, RVs, linear algebra, gradient descent.

Same conclusion from five different lenses — that's what makes the result trustworthy.

— END OF PRESENTATION —



Thanks *a latte.*

QUESTIONS · DISCUSSION